

Exercise 1.1 Adapt the proof showing that $H(X) \leq \log |\mathcal{X}|$ for binary random variables so that it works for general finite alphabets $\mathcal{X}$.

Solution: Let us define $f(t) = -t \log t$, which is strictly concave in t .

$$\begin{aligned}
 H(X) &= - \sum_{x \in \mathcal{X}} p(x) \log p(x) \\
 &= \sum_{x \in \mathcal{X}} f(p(x)) \\
 &= |\mathcal{X}| \sum_{x \in \mathcal{X}} \frac{1}{|\mathcal{X}|} f(p(x)) \\
 &\leq |\mathcal{X}| f\left(\sum_{x \in \mathcal{X}} \frac{1}{|\mathcal{X}|} p(x)\right) \\
 &= |\mathcal{X}| f\left(\frac{1}{|\mathcal{X}|}\right) \\
 &= \log |\mathcal{X}|,
 \end{aligned}$$

where the inequality follows from the concavity of f .

Exercise 1.2 For each item, find an example of random variables X , Y and Z such that the desired relations holds:

- a) $H(X|YZ) = 0$ and $H(X|Y) = H(X|Z) = 1$.
- b) $I(X : Y|Z) = 1$ and $I(X : Y) = 0$.
- c) $I(X : Y) = 1$ and $I(X : Y|Z) = 0$.
- d) $I(X : Y) = I(X : Z) = 1$ and $I(Y : Z) = 0$.

Hint: An alphabet of one or two bits for each random variable will suffice.

Solution:

- a.) We may choose binary random variables satisfying $X = Y \oplus Z$ with Y and Z uniform and independent.
- b.) We may choose X and Y uniform and independent with $Z = X \oplus Y$.
- c.) We may choose binary $X = Y = Z$ uniform.
- d.) We may choose $X = (Y, Z)$ with Y and Z independent and uniform.

Exercise 1.3 Compute the mutual information $I(X : Y)$ for the pmf

P_{XY}	$x = 0$	$x = 1$	$x = 2$
$y = 0$	$\frac{1}{2}$	0	0
$y = 1$	0	$\frac{1}{4}$	$\frac{1}{4}$

Solution: We can conclude immediately from the table that the pmf of Y is

$$P_Y(y) = \frac{1}{2}, \quad y \in \{0, 1\},$$

and the conditional pmf of Y given X is

$$P_{Y|X}(y|x) = \begin{cases} \delta_{y,0}, & x \in \{0\}, \\ \delta_{y,1}, & x \in \{1, 2\}. \end{cases}$$

Therefore, the entropy of Y is

$$H(Y) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = 1,$$

and the conditional entropy of Y given X is

$$H(Y|X) = -\frac{1}{2} \log 1 - \frac{1}{4} \log 1 - \frac{1}{4} \log 1 = 0.$$

As a result, the mutual information is given by

$$I(X : Y) = H(Y) - H(Y|X) = 1.$$

Exercise 1.4 Consider two sequences of random variables $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$. Show that if $X_1, \dots, X_n$ are mutually independent, then

$$I(X_1, \dots, X_n : Y_1, \dots, Y_n) \geq \sum_{i=1}^n I(X_i : Y_i).$$

On other hand, if given Y_i the random variable X_i is conditionally independent of all the other random variables for all $i = 1, \dots, n$, then show that

$$I(X_1, \dots, X_n : Y_1, \dots, Y_n) \leq \sum_{i=1}^n I(X_i : Y_i).$$

Solution: For the first claim, consider

$$\begin{aligned} I(X_1, \dots, X_n : Y_1, \dots, Y_n) &= \sum_{i=1}^n I(X_i : Y_1, \dots, Y_n | X_1, \dots, X_{i-1}) \\ &= \sum_{i=1}^n I(X_i : Y_1, \dots, Y_n, X_1, \dots, X_{i-1}) - I(X_i : X_1, \dots, X_{i-1}) \\ &= \sum_{i=1}^n I(X_i : Y_1, \dots, Y_n, X_1, \dots, X_{i-1}) \\ &\geq \sum_{i=1}^n I(X_i : Y_i), \end{aligned}$$

where the third equality is by independence.

For the second claim, consider

$$\begin{aligned}
I(X_1, \dots, X_n : Y_1, \dots, Y_n) &= H(X_1, \dots, X_n) - H(X_1, \dots, X_n | Y_1, \dots, Y_n) \\
&= H(X_1, \dots, X_n) - \sum_{i=1}^n H(X_i | Y_1, \dots, Y_n, X_1, \dots, X_{i-1}) \\
&= H(X_1, \dots, X_n) - \sum_{i=1}^n H(X_i | Y_i) \\
&\leq \sum_{i=1}^n H(X_i) - \sum_{i=1}^n H(X_i | Y_i) \\
&= \sum_{i=1}^n I(X_i : Y_i),
\end{aligned}$$

where the third equality is by the fact that given Y_i , X_i is conditionally independent of all other random variables for $i = 1, \dots, n$ so

$$H(X_i | Y_1, \dots, Y_n, X_1, \dots, X_{i-1}) = H(X_i | Y_i).$$

Exercise 1.5 (Parent quantity) Let X and Y be random variables on alphabets $\mathcal{X}$ and $\mathcal{Y}$ with joint pmf P_{XY} . Moreover, let U be a uniform random variable on $\mathcal{X}$. Show the following relations:

- a) $H(X) = \log |\mathcal{X}| - D(P_X \| U_X)$.
- b) $H(X|Y) = \log |\mathcal{X}| - D(P_{XY} \| U_X \times P_Y)$.
- c) $I(X : Y) = D(P_{XY} \| P_X \times P_Y)$.

Solution:

a.) By definition,

$$\begin{aligned}
H(X) &= - \sum_{x \in \mathcal{X}} P_X(x) \log P_X(x) \\
&= - \sum_{x \in \mathcal{X}} P_X(x) \left(\log \frac{P_X(x)}{1/|\mathcal{X}|} + \log \frac{1}{|\mathcal{X}|} \right) \\
&= \log |\mathcal{X}| - D(P_X \| U_X).
\end{aligned}$$

b.) By definition,

$$\begin{aligned}
H(X|Y) &= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x, y) \log P_{X|Y}(x|y) \\
&= - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x, y) \left(\log \frac{P_{XY}(x, y)}{P_Y(y)/|\mathcal{X}|} + \log \frac{1}{|\mathcal{X}|} \right) \\
&= \log |\mathcal{X}| - D(P_{XY} \| U_X \times P_Y).
\end{aligned}$$

c.) By definition,

$$\begin{aligned}
I(X : Y) &= H(Y) - H(Y|X) = \sum_{y \in \mathcal{Y}} P_Y(y) \log \frac{1}{P_Y(y)} - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x, y) \log \frac{P_X(x)}{P_{XY}(x, y)} \\
&= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x, y) \log \frac{1}{P_Y(y)} - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x, y) \log \frac{P_X(x)}{P_{XY}(x, y)} \\
&= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P_{XY}(x, y) \log \frac{P_{XY}(x, y)}{P_X(x)P_Y(y)} \\
&= D(P_{XY} \| P_X \times P_Y).
\end{aligned}$$

Exercise 1.6 (Information Spectrum) Given a random variable X governed by the pmf P or an alternative pmf Q , recall that the log-likelihood ratio is defined as the random variable $Z(X) = \log \frac{P(X)}{Q(X)}$ where X is distributed according to P .

- a.) Give an expression for $\text{Var}[Z]$ (under P). This quantity is called the relative entropy variance and denoted by $V(P\|Q)$.

Consider now a sequence of i.i.d. random variables $X^n = (X_1, X_2, \dots, X_n)$ on $\mathcal{X}^n$ where each X_i is governed by the pmf P or an alternative pmf Q . We are interested in pmf of the log-likelihood ratio $Z(X^n)$.

- b.) Show that $Z(X^n) = \sum_{i=1}^n Z(X_i)$. What is $\mathbb{E}[Z^n]$ and $\text{Var}[Z^n]$?

- c.) Let us now consider the quantity $P[Z(X^n) \leq nR]$ in the limit of large n for different values of R . Show that

$$\lim_{n \rightarrow \infty} P[Z(X^n) \leq nR] = \begin{cases} 0 & \text{if } R < D(P\|Q) \\ 1 & \text{if } R > D(P\|Q) \end{cases}.$$

Hint: Argue using the weak law of large numbers.

- d.) Later on in the lecture we will encounter the quantity

$$D_s^\epsilon(P^n\|Q^n) := \sup\{k \in \mathbb{R} : P[Z(X^n) \leq k] \leq \epsilon\}, \quad (1)$$

which, in words, is the largest k such that the tail of the distribution of Z that lies below k has cumulative probability at most ϵ . Show that $D_s^\epsilon(P^n\|Q^n) = nD(P\|Q) + o(n)$, or equivalently,

$$\lim_{n \rightarrow \infty} \frac{1}{n} D_s^\epsilon(P^n\|Q^n) = D(P\|Q).$$

Hint: Verify that $\frac{1}{n} D_s^\epsilon(P^n\|Q^n) = \sup\{k \in \mathbb{R} : P[\frac{1}{n} Z(X^n) \leq k] \leq \epsilon\}$.

- e.) Optional: Show that in the next order in n , we have

$$D_s^\epsilon(P^n\|Q^n) = nD(P\|Q) + \sqrt{nV(P\|Q)} \Phi^{-1}(\epsilon) + o(\sqrt{n})$$

Can we even say something more about the $o(\sqrt{n})$ term?

Hint: The statement can be shown using the central limit theorem. A quantitative version of the central limit theorem is the Berry–Esseen theorem. Look it up to make even stronger statements about the remainder term.

Solution:

- a.)

$$V(P\|Q) = \text{Var}[Z] = \sum_{x \in \mathcal{X}} P(x) \left(\log \frac{P(x)}{Q(x)} - D(P\|Q) \right)^2.$$

- b.)

$$\begin{aligned} Z(X^n) &= \log \frac{P(X^n)}{Q(X^n)} = \log \frac{\prod_{i=1}^n P(X_i)}{\prod_{i=1}^n Q(X_i)} = \sum_{i=1}^n \log \frac{P(X_i)}{Q(X_i)} = \sum_{i=1}^n Z(X_i), \\ \mathbb{E}[Z^n] &= \sum_{i=1}^n \mathbb{E}[Z(X_i)] = nD(P\|Q). \end{aligned}$$

We use independence to write

$$\text{Var}[Z^n] = \sum_{i=1}^n \text{Var}[Z(X_i)] = nV(P\|Q).$$

c) Using the weak law of large numbers, we have for any positive number $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} P \left[\left| \frac{\sum_{i=1}^n Z(X_i)}{n} - D(P\|Q) \right| > \epsilon \right] = 0,$$

and

$$\lim_{n \rightarrow \infty} P \left[\left| \frac{\sum_{i=1}^n Z(X_i)}{n} - D(P\|Q) \right| \leq \epsilon \right] = 1.$$

Then, in particular,

$$\begin{aligned} \lim_{n \rightarrow \infty} P[Z(X^n) \leq n(D(P\|Q) - \epsilon)] &= 0, \\ \lim_{n \rightarrow \infty} P[Z(X^n) \leq n(D(P\|Q) + \epsilon)] &= 1. \end{aligned}$$

Now, if $R < D(P\|Q)$ then there also exists an $\epsilon > 0$ such that $R \leq D(P\|Q) - \epsilon$. And similalry, if $R > D(P\|Q)$ then there exists an $\epsilon > 0$ such that $R \geq D(P\|Q) + \epsilon$. Hence, the above inequalities imply the desired result.

d) From the previous item and the definition of the limit we know that for any $R < D(P\|Q)$ and $\epsilon > 0$, there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ we have

$$P[Z(X^n) \leq nR] \leq \epsilon.$$

This implies that in the definition of (1) we are allowed to choose any $k \leq nR$ and thus by taking the supremum we get

$$D_s^\epsilon(P^n\|Q^n) \geq \sup\{k \in \mathbb{R} : k < nR\} = nR.$$

Taking the limit $n \rightarrow \infty$ yields the desired lower bound, for all $R < D(P\|Q)$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} D_s^\epsilon(P^n\|Q^n) \geq R,$$

And hence $\lim_{n \rightarrow \infty} \frac{1}{n} D_s^\epsilon(P^n\|Q^n) \geq D(P\|Q)$ since this holds for all $R < D(P\|Q)$.

We can argue similarly in the opposite direction. For any $R > D(P\|Q)$ and $\mu > 0$, there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ we have

$$P[Z(X^n) \leq nR] \geq 1 - \mu.$$

If we choose μ small enough so that $1 - \mu > \epsilon$ then this implies that any $k \geq nR$ violates the constraint on the probability in the definition of $D_s^\epsilon(P^n\|Q^n)$, and thus we must have

$$D_s^\epsilon(P^n\|Q^n) \leq nR.$$

Taking again the limit we find that

$$\lim_{n \rightarrow \infty} \frac{1}{n} D_s^\epsilon(P^n\|Q^n) \leq R,$$

Since this holds for any $R > D(P\|Q)$ we deduce that $\lim_{n \rightarrow \infty} \frac{1}{n} D_s^\epsilon(P^n\|Q^n) \leq D(P\|Q)$.

e) We give here actually an even stronger bound, using the Berry–Esseen theorem, which tells us how quickly the renormalized distribution approaches the Gaussian distribution in the central limit theorem. To make this precise we define

$$T(P\|Q) := \sum_{x \in \mathcal{X}} P(x) \left| \log \frac{P(x)}{Q(x)} - D(P\|Q) \right|^3.$$

By the Berry-Esseen theorem, we have

$$\left| P[Z(X^n) \leq k] - \Phi\left(\frac{\sqrt{n}(k/n - D(P\|Q))}{\sqrt{V(P\|Q)}}\right) \right| \leq \frac{6T(P\|Q)}{\sqrt{nV^3(P\|Q)}}.$$

By letting

$$\Phi\left(\frac{\sqrt{n}(k/n - D(P\|Q))}{\sqrt{V(P\|Q)}}\right) + \frac{6T(P\|Q)}{\sqrt{nV^3(P\|Q)}} \geq \epsilon$$

and

$$\Phi\left(\frac{\sqrt{n}(k/n - D(P\|Q))}{\sqrt{V(P\|Q)}}\right) - \frac{6T(P\|Q)}{\sqrt{nV^3(P\|Q)}} \leq \epsilon,$$

we can constrain k as follows:

$$\begin{aligned} nD(P\|Q) + \sqrt{nV(P\|Q)}\Phi^{-1}\left(\epsilon - \frac{6T(P\|Q)}{\sqrt{nV^3(P\|Q)}}\right) &\leq k \\ &\leq nD(P\|Q) + \sqrt{nV(P\|Q)}\Phi^{-1}\left(\epsilon + \frac{6T(P\|Q)}{\sqrt{nV^3(P\|Q)}}\right). \end{aligned}$$

This implies the two bounds

$$\begin{aligned} D_s^\epsilon(P^n\|Q^n) &\geq nD(P\|Q) + \sqrt{nV(P\|Q)}\Phi^{-1}\left(\epsilon - \frac{6T(P\|Q)}{\sqrt{nV^3(P\|Q)}}\right) \\ D_s^\epsilon(P^n\|Q^n) &\leq nD(P\|Q) + \sqrt{nV(P\|Q)}\Phi^{-1}\left(\epsilon + \frac{6T(P\|Q)}{\sqrt{nV^3(P\|Q)}}\right) \end{aligned}$$

Equivalently,

$$\lim_{n \rightarrow \infty} \frac{1}{n} D_s^\epsilon(P^n\|Q^n) = D(P\|Q).$$

If $V(P\|Q) > 0$ and $T(P\|Q) < \infty$, the term $\frac{6T(P\|Q)}{\sqrt{nV^3(P\|Q)}}$ is equal to $c/\sqrt{n}$ for some finite $c > 0$. By Taylor expansions,

$$\Phi^{-1}\left(\epsilon \pm \frac{c}{\sqrt{n}}\right) = \Phi^{-1}(\epsilon) + O\left(\frac{1}{\sqrt{n}}\right).$$

By plugging in the Taylor expansion, we can get the result.